

Report
Of
TATA Cybersecurity Analyst Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
Roll Number of Student: 25SCS1003000765
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Batch: 2025-29

2026-27

CANDIDATE'S DECLARATION

I, Raghav Bhujel do hereby declare that the internship report titled **“Cybersecurity Analyst”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name Raghav Bhujel

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

Student Name Raghav Bhujel

Roll No.. 25SCS1003000765

Internship / Job Simulation Completion Certificate



Inspiring and
empowering
future professionals

Raghav Bhujel Cybersecurity Analyst Job Simulation

Certificate of Completion
September 9th, 2026

Over the period of September 2026, Raghav Bhujel has completed practical tasks in:

Identity and access management (IAM) fundamentals
IAM strategy assessment
Crafting custom IAM solutions
Platform integration

Tom Brunskill
Co-Founder of Forage

Enrollment Verification Code: 6a000a73a9a084a0f90751a4d | User Verification Code: 6a000a73a9a084a0f90751a4d | Issued by Forage

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1. Introduction to the Job Simulation

This report documents the TATA Cybersecurity Analyst Virtual Experience Program hosted on the Forage platform. The program simulates the responsibilities of an IAM Analyst working with TechCorp, a fictional enterprise undergoing digital transformation. The objective is to understand enterprise IAM practices and prepare implementation strategies for secure identity management.

2. Organization Profile – Tata Consultancy Services (TCS)

TCS is a global IT services and consulting company providing cybersecurity, cloud, digital transformation, and identity security services. TCS IAM services include readiness assessment, IAM solution design, IAM platform implementation, SSO integration, access governance, Identity-as-a-Service, and managed IAM services.

3. Problem Statement

TechCorp requires a secure Identity and Access Management framework to protect enterprise applications, reduce unauthorized access, comply with regulations, and improve user authentication while supporting business growth.

4. Project Objectives

- Understand IAM fundamentals and enterprise identity security.
- Assess IAM readiness of TechCorp.
- Design customized IAM strategies using Least Privilege, RBAC, MFA, and lifecycle management.
- Prepare an IAM implementation and integration roadmap.
- Improve security while maintaining user experience through SSO and standardized authentication protocols.

5. Scope of the Project

The scope includes IAM readiness assessment, identity verification, access governance, role-based access control, user lifecycle management, compliance monitoring, IAM integration with enterprise applications, testing, deployment, and continuous monitoring.

6. Technologies and Tools Used

Identity and Access Management concepts, RBAC, MFA, OAuth 2.0, SAML, Single Sign-On (SSO), Identity-as-a-Service (IDaaS), Access Governance, Audit Logging, Compliance Frameworks, and IAM Platform Implementation methodology.

Table 6.1 Technologies and Their Purpose

Technology	Purpose
IAM	Manage digital identities and access permissions
RBAC	Assign permissions based on organizational roles
MFA	Strengthen authentication security
OAuth 2.0	Secure delegated authorization
SAML	Enterprise authentication and federation
SSO	Single authentication for multiple applications
Audit & Monitoring	Track user activities and compliance

7. System Architecture

TechCorp's IAM architecture follows a centralized identity provider connected to enterprise applications through SSO, RBAC, MFA, audit logging, and lifecycle management. The architecture ensures authentication, authorization, monitoring, and secure access across cloud and on-premises resources.

8. Methodology

- Project Initiation: Define IAM scope, stakeholders, and objectives.
- Needs Assessment: Review existing infrastructure and identify IAM gaps.
- Solution Design: Configure RBAC, least privilege, MFA, and lifecycle management.
- Resource Planning: Allocate IAM resources and integration requirements.
- Implementation: Configure IAM platform and authentication services.
- Testing & QA: Validate authentication, authorization, and security controls.
- Deployment: Roll out IAM implementation in phases.
- Monitoring & Optimization: Continuous monitoring, auditing, and security improvement.

Project Deliverables

Task	Description
IAM Fundamentals	Understanding identity verification, access control, complia
IAM Strategy Assessment	Assessing readiness and designing customized IAM strate
Crafting IAM Solutions	Applying least privilege, RBAC, MFA, lifecycle managemen
Platform Integration	Implementation roadmap and integration best practices.

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9. Expected Outcomes

Expected outcomes of the job simulation include understanding enterprise IAM architecture, designing IAM strategies aligned with business objectives, improving authentication through MFA and SSO, implementing RBAC, ensuring compliance through audit logging, and preparing an implementation roadmap for a secure IAM platform.

Skills Learned

Technical Skills	Professional Skills
IAM Fundamentals	Security Analysis
RBAC and Least Privilege	Problem Solving
MFA, OAuth 2.0, SAML	Documentation
Single Sign-On (SSO)	Communication
Access Governance & Compliance	Analytical Thinking
User Lifecycle Management	Stakeholder Understanding

10. Conclusion

The TATA Cybersecurity Analyst Virtual Experience Program provided practical exposure to enterprise Identity and Access Management concepts using the TechCorp case study. The simulation demonstrated how IAM improves cybersecurity through identity verification, least privilege access, RBAC, MFA, compliance auditing, and secure application integration. The project strengthened understanding of IAM implementation planning and enterprise cybersecurity strategy.

11. Bibliography / References

- Tata Consultancy Services – Identity and Access Management (IAM) Services.
- Forage – TATA Cybersecurity Analyst Virtual Experience Program.
- OAuth 2.0 Authorization Framework documentation.
- SAML 2.0 Security Assertion Markup Language documentation.
- Identity and Access Management best practices for enterprise cybersecurity.